

CS215 Assignment-1

Malay Kedia (Roll No. 23B1034)
Aakash Gupta (Roll No. 23B0953)
Yash Singh (Roll No. 23B0934)

1. A Pretty “Normal” Mixture

Task A:

Solution:

$$\begin{aligned} F_A(c) &= P(A < c) \\ &= \sum_{i=1}^K P(A < c \wedge A = X_i) && \text{(By law of total probability)} \\ &= \sum_{i=1}^K P(A < c \mid A = X_i) \cdot P(A = X_i) \\ &= \sum_{i=1}^K P(X_i < c) \cdot P(A = X_i) \\ &= \sum_{i=1}^K p_i F_{X_i}(c) \end{aligned}$$

(Substituting the CDF $F_{X_i}(c)$ for each Gaussian component)

To find the probability density function (PDF) $f_A(c)$, we differentiate $F_A(c)$ with respect to c .

$$\begin{aligned} f_A(c) &= \frac{d}{dc} F_A(c) \\ &= \sum_{i=1}^K p_i \cdot \frac{d}{dc} F_{X_i}(c) \\ &= \sum_{i=1}^K p_i \cdot f_{X_i}(c) && \left(\frac{d}{dc} F_{X_i}(c) = f_{X_i}(c) \right) \end{aligned}$$

The PDF of the GMM variable X is f_X , given by $f_X(x) = \sum_{i=1}^K p_i \cdot f_{X_i}(x)$, which is exactly same as the one I got from my algorithm. Hence, $\forall u \in \mathbb{R}, f_A(u) = f_X(u)$.

Task B:

Solution:

Part 3:

$$\begin{aligned}M_X(t) &= E(e^{tX}) \\&= \int_{-\infty}^{\infty} e^{tx} f_X(x) dx \\&= \int_{-\infty}^{\infty} e^{tx} \cdot \left(\sum_{i=1}^K p_i \cdot f_{X_i}(x) \right) dx \\&= \sum_{i=1}^K \left(p_i \int_{-\infty}^{\infty} e^{tx} \cdot f_{X_i}(x) dx \right) \\&= \sum_{i=1}^K p_i E(e^{tX_i}) \\&= \sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}}\end{aligned}$$

We can now find the expected value $E[X]$ and variance $Var[X]$ using the MGF.

Part 1:

Differentiating the MGF with respect to t :

$$\begin{aligned}M'_X(t) &= \frac{d}{dt} \left(\sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \right) \\&= \sum_{i=1}^K p_i \cdot \frac{d}{dt} \left(e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \right) \\&= \sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \cdot (\mu_i + \sigma_i^2 t)\end{aligned}$$

Now, $E[X] = M'_X(0)$

$$\begin{aligned}E[X] &= M'_X(0) = \sum_{i=1}^K p_i \cdot e^0 \cdot \mu_i \\&= \sum_{i=1}^K p_i \cdot \mu_i\end{aligned}$$

Part 2:

$$\begin{aligned} M_X''(t) &= \sum_{i=1}^K p_i \cdot \frac{d^2}{dt^2} \left(e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \right) \\ &= \sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \cdot (\mu_i + \sigma_i^2 t)^2 + \sum_{i=1}^K p_i \cdot e^{\mu_i t + \frac{\sigma_i^2 t^2}{2}} \cdot \sigma_i^2 \end{aligned}$$

Now, the second moment $E[X^2]$ is:

$$E[X^2] = M_X''(0) = \sum_{i=1}^K p_i \cdot (\mu_i^2 + \sigma_i^2)$$

Now, the variance $\text{Var}(X)$ is:

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ \text{Var}(X) &= \sum_{i=1}^K p_i \cdot (\mu_i^2 + \sigma_i^2) - \left(\sum_{i=1}^K p_i \cdot \mu_i \right)^2 \end{aligned}$$

Task C:

Solution:

ATQ, $Z = \sum_{i=1}^K p_i X_i$.

Part 4:

$$\begin{aligned} M_Z(t) &= E[e^{tZ}] \\ &= E[e^{t \sum_{i=1}^K p_i X_i}] \\ &= E\left[\prod_{i=1}^K e^{t p_i X_i}\right] \\ &= \prod_{i=1}^K E[e^{t p_i X_i}] && \text{(As variables are independent)} \\ &= \prod_{i=1}^K e^{\mu_i p_i t + \frac{\sigma_i^2 p_i^2 t^2}{2}} \\ &= e^{\sum_{i=1}^K \mu_i p_i t + \frac{\sigma_i^2 p_i^2 t^2}{2}} \\ &= e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \end{aligned}$$

Part 1:

Differentiating $M_Z(t)$ with respect to t :

$$\begin{aligned} M'_Z(t) &= \frac{d}{dt} \left(e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \right) \\ &= e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \cdot \left(\sum_{i=1}^K \mu_i p_i + t \sum_{i=1}^K \sigma_i^2 p_i^2 \right) \end{aligned}$$

$$\begin{aligned} E[Z] &= M'_Z(0) = e^0 \cdot \left(\sum_{i=1}^K \mu_i p_i \right) \\ &= \sum_{i=1}^K \mu_i p_i \end{aligned}$$

Part 2:

$$\begin{aligned} M''_Z(t) &= \frac{d}{dt} \left(e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \cdot \left(\sum_{i=1}^K \mu_i p_i + t \sum_{i=1}^K \sigma_i^2 p_i^2 \right) \right) \\ &= e^{t \sum_{i=1}^K \mu_i p_i + \frac{t^2 \sum_{i=1}^K \sigma_i^2 p_i^2}{2}} \cdot \left(\left(\sum_{i=1}^K \mu_i p_i + t \sum_{i=1}^K \sigma_i^2 p_i^2 \right)^2 + \sum_{i=1}^K \sigma_i^2 p_i^2 \right) \end{aligned}$$

Evaluating at 0,

$$E[Z^2] = M''_Z(0) = e^0 \cdot \left(\left(\sum_{i=1}^K \mu_i p_i \right)^2 + \sum_{i=1}^K \sigma_i^2 p_i^2 \right) = \left(\sum_{i=1}^K \mu_i p_i \right)^2 + \sum_{i=1}^K \sigma_i^2 p_i^2$$

Now, the variance is:

$$\begin{aligned} \text{Var}(Z) &= E[Z^2] - (E[Z])^2 \\ &= \left(\sum_{i=1}^K \mu_i p_i \right)^2 + \sum_{i=1}^K \sigma_i^2 p_i^2 - \left(\sum_{i=1}^K \mu_i p_i \right)^2 \\ &= \sum_{i=1}^K \sigma_i^2 p_i^2 \end{aligned}$$

Part 3: PDF of Z

We observe that the MGF of Z matches the MGF of the gaussian $\mathcal{N}(\mu, \sigma^2)$ with $\mu = \sum_{i=1}^K \mu_i p_i$ and $\sigma^2 = \sum_{i=1}^K \sigma_i^2 p_i^2$.

$$Z \sim \mathcal{N} \left(\sum_{i=1}^K \mu_i p_i, \sum_{i=1}^K \sigma_i^2 p_i^2 \right)$$

Thus, the PDF of Z is:

$$f_Z(u) = \frac{1}{\sqrt{2\pi \sum_{i=1}^K \sigma_i^2 p_i^2}} \exp \left(-\frac{(u - \sum_{i=1}^K \mu_i p_i)^2}{2 \sum_{i=1}^K \sigma_i^2 p_i^2} \right)$$

Part 5:

No, X and Z have very different behaviors, and don't even have the same variances, much less the PDFs.

Part 6:

As concluded in part 3, Z is a gaussian.

Task D:

Solution:

Let X be a finite discrete random variable taking values $x_1, x_2, \dots, x_n$ with respective probabilities $P(X = x_i) = p_i$, for $i = 1, 2, \dots, n$.

The moment generating function (MGF) of X , denoted by $\phi_X(t)$, is given by:

$$\phi_X(t) = E[e^{tX}] = \sum_{i=1}^n e^{tx_i} P(X = x_i) = \sum_{i=1}^n e^{tx_i} p_i$$

By the above construction, it is trivial to show that if PMF of two random variables is same, their MGF is the same. Hence, the PMF uniquely determines the MGF.

Given the MGF $\phi_X(t) = \sum_{i=1}^n e^{tx_i} p_i$, we can determine the probability mass function (PMF) by solving a system of linear equations. If we know $\phi_X(t)$ for enough values of t , we can solve for the probabilities $p_1, p_2, \dots, p_n$ by inverting this system. Therefore, the MGF uniquely determines the PMF.

Conclusion about X and Z

Since Z is a weighted sum of independent Gaussian random variables and does not have the same MGF as X , they don't have the same distribution (by the theorem). This is also the expected logical outcome, as X is a GMM, in which probability of the rv itself is a weighted sum of probabilities of n gaussian random variables, while Z is the sum of outcomes of n independent gaussian random variables, which turns out of itself be a gaussian.